

Cross-Domain Method Transfer Study v1

Does a method’s in-domain track record carry to a neighboring domain?

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Abstract

We ask whether a method with a strong, large- n in-domain track record transfers to a neighboring domain it has no track record in. The method under study is a domain-specialist coherence classifier — a multinomial logistic regression over nine geometry features of a (premise, continuation) embedding pair — trained on the frozen Quintin Hypothesis v1 dataset. We evaluate 3 method/domain pairs against frozen, hash-pinned held-out eval sets in the neighboring domains chemistry, finance, and law. For each pair we report in-domain cross-validated accuracy, transfer accuracy, a domain-naïve baseline trained directly on the target domain, an unpaired bootstrap confidence interval on the in-domain-minus-transfer accuracy gap, a two-proportion z -test, and Cohen’s h . Across 3 method/domain pair(s): 1 partial transfer; 2 no transfer. The firm publishes whichever it finds — partial-transfer cases are reported with the specific sub-capability that survives the domain boundary, losses included.

1 Setup

The source domain D supplies the method’s track record: the method is trained on the frozen QH v1 slice for D , and its in-domain accuracy is the 5-fold cross-validated, out-of-fold accuracy on that slice. The target domain D' is a neighboring domain the method has no resolved items in. Transfer accuracy is the D -trained weights applied to the frozen D' eval set, with the feature scaler re-fit on D' (unsupervised — no D' labels are used); the strict zero-adaptation variant is reported alongside. The domain-naïve baseline is the same architecture trained directly on D' by cross-validation. Every dataset is hash-pinned in the pairs manifest (`benchmarks/transfer/v1/pairs.yaml`) and re-verified on each run; the held-out sets are frozen and not re-curated to chase a result. Run envelope: git SHA 0034929158a42e4e536d85efd41ab22721c7ca50, embedder `hash-det-v1`, bootstrap 10000 resamples, seed 17.

2 Results

Pair	In-dom.	Transfer	Baseline	Δ 95% CI	Cohen’s h	Verdict
physics-to-chemistry	0.688	0.227	0.744	[0.393, 0.528]	0.963	No transfer
economics-to-finance	0.695	0.494	0.463	[0.116, 0.286]	0.413	Partial transfer
ethics-to-law	0.427	0.351	0.491	[-0.026, 0.176]	0.157	No transfer

2.1 physics-to-chemistry: physics $\rightarrow$ chemistry

Chemistry is the natural-science neighbor of physics: same template construction (parameterized premise, near-identical coherent / contradicting continuations, off-topic orthogonal), different domain vocabulary. The method has no QH track record in chemistry and declares no DomainBound naming it.

- In-domain track record: accuracy 0.6876 ($n = 1018$), Brier 0.2159, ECE 0.1558.
- Transfer to chemistry: accuracy 0.2267 ($n = 172$), Brier 0.4210, ECE 0.4600. Strict zero-adaptation variant: 0.2849.
- Sub-capability split: orthogonal-vs-rest 0.5174, coherent-vs-contradicting 0.1262.
- Domain-naive baseline trained on chemistry: accuracy 0.7442.
- In-domain minus transfer gap 0.4609, 95% bootstrap CI [0.3927, 0.5277], bootstrap $p = 0.0000$; two-proportion $z = 11.5236$, $p = 0.0000$; Cohen's $h = 0.9629$ (large).
- Transfer vs. chance (0.333): one-sided $z = -2.9654$, $p = 0.9985$.

Verdict — No transfer. transfer accuracy is not significantly above random chance (0.333) — the method's specialization does not carry into this domain.

2.2 economics-to-finance: economics $\rightarrow$ finance

Finance is the quantitative-reasoning neighbor of economics: shared structural intuitions (prices, rates, risk) over a distinct vocabulary. No QH track record in finance; no DomainBound naming it.

- In-domain track record: accuracy 0.6951 ($n = 574$), Brier 0.2389, ECE 0.0693.
- Transfer to finance: accuracy 0.4938 ($n = 160$), Brier 0.2421, ECE 0.1076. Strict zero-adaptation variant: 0.3688.
- Sub-capability split: orthogonal-vs-rest 0.6750, coherent-vs-contradicting 0.4405.
- Domain-naive baseline trained on finance: accuracy 0.4625.
- In-domain minus transfer gap 0.2014, 95% bootstrap CI [0.1155, 0.2861], bootstrap $p = 0.0000$; two-proportion $z = 4.7264$, $p = 0.0000$; Cohen's $h = 0.4134$ (small).
- Transfer vs. chance (0.333): one-sided $z = 4.3044$, $p = 0.0000$.

Verdict — Partial transfer. transfer accuracy is significantly above chance but also significantly worse than in-domain — the capability carries over only partially.

2.3 ethics-to-law: ethics $\rightarrow$ law

Law is the normative-reasoning neighbor of ethics: both reason over rules, agents, and permissibility, with a distinct vocabulary. No QH track record in law; no DomainBound naming it.

- In-domain track record: accuracy 0.4273 ($n = 344$), Brier 0.2386, ECE 0.1065.

- Transfer to law: accuracy 0.3509 ($n = 114$), Brier 0.2857, ECE 0.1975. Strict zero-adaptation variant: 0.2807.
- Sub-capability split: orthogonal-vs-rest 0.5702, coherent-vs-contradicting 0.3846.
- Domain-naive baseline trained on law: accuracy 0.4912.
- In-domain minus transfer gap 0.0764, 95% bootstrap CI [-0.0258, 0.1758], bootstrap $p = 0.1298$; two-proportion $z = 1.4392$, $p = 0.1501$; Cohen’s $h = 0.1570$ (negligible).
- Transfer vs. chance (0.333): one-sided $z = 0.3974$, $p = 0.3456$.

Verdict — No transfer. transfer accuracy is not significantly above random chance (0.333) — the method’s specialization does not carry into this domain.

3 Honest findings

The study exists so the firm can publish a method that does not generalize. The losses, stated plainly:

- physics-to-chemistry: NO TRANSFER. In-domain accuracy 0.6876 collapses to 0.2267 on chemistry — a significant drop, and not significantly above the 0.333 chance floor. The method’s specialization in physics does not carry into the neighboring domain.
- physics-to-chemistry: a domain-naive baseline *trained on chemistry* (0.7442) beats the transferred method (0.2267) — fitting the new domain directly would have done better than carrying the specialist over.
- physics-to-chemistry: covariate shift is load-bearing — the strict zero-adaptation transfer scores 0.2849 vs 0.2267 once the feature scaler is re-fit on chemistry (unsupervised).
- economics-to-finance: PARTIAL TRANSFER. In-domain accuracy 0.6951 degrades to 0.4938 on finance — a significant loss, but still above chance. Sub-capability split: orthogonal-vs-rest 0.6750, coherent-vs-contradicting 0.4405.
- economics-to-finance: covariate shift is load-bearing — the strict zero-adaptation transfer scores 0.3688 vs 0.4938 once the feature scaler is re-fit on finance (unsupervised).
- ethics-to-law: NO TRANSFER. In-domain accuracy is already a modest 0.4273, and transfer accuracy 0.3509 on law is not significantly above the 0.333 chance floor. The method’s specialization in ethics does not carry into the neighboring domain.
- ethics-to-law: a domain-naive baseline *trained on law* (0.4912) beats the transferred method (0.3509) — fitting the new domain directly would have done better than carrying the specialist over.
- ethics-to-law: covariate shift is load-bearing — the strict zero-adaptation transfer scores 0.2807 vs 0.3509 once the feature scaler is re-fit on law (unsupervised).

4 What this study does not do

Per the study constraints, this experiment does not modify any method’s declared domain bound (`noosphere/methods/domain_bounds.py`). Widening or narrowing a method’s declared domain is a founder decision that follows the published evidence; it is not a side effect of the experiment. Every number above is produced by `noosphere.transfer.study` from the frozen, hash-pinned inputs — no value is hand-edited.